ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

# ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

## PRACTICE EXAM 4 · MARKING GUIDELINE

**MARKS: 80**

**Marking guideline for the Academic Central NC(V) Level 2 Electronics and Digital Concepts for Robotics practice exam 4.**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

**How to mark:** each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers, equivalent designs and neat, recognisable diagrams.

## SECTION A

## QUESTION 1

**1.1** 1.1.1 A ✓ 1.1.2 C ✓ 1.1.3 B ✓ 1.1.4 B ✓ 1.1.5 A ✓ 1.1.6 D ✓ 1.1.7 C ✓ 1.1.8 A ✓ 1.1.9 B ✓ 1.1.10 D ✓ (10)

**1.2** 1.2.1 C ✓ 1.2.2 J ✓ 1.2.3 I ✓ 1.2.4 F ✓ 1.2.5 E ✓ 1.2.6 G ✓ 1.2.7 H ✓ 1.2.8 D ✓ 1.2.9 B ✓ 1.2.10 A ✓ (10)

**TOTAL SECTION A: [20]**

## SECTION B

## QUESTION 2

**2.1.1** $\mathrm{R} = (\mathrm{V_s} - \mathrm{V_{LED}}) \div \mathrm{I}$ ✓ $= (6 - 3{,}2) \div 0{,}02 = 140\ \Omega$ ✓; choose $150\ \Omega$ ✓ (3)

**2.1.2** $\mathrm{I_T} = 20 + 20 = 40\ \mathrm{mA}$ ✓ (1)

**2.1.3** $\mathrm{P} = \mathrm{I}^2\mathrm{R}$ ✓ $= 0{,}02^2 \times 140 = 0{,}06\ \mathrm{W}$ ✓ (2)

**2.2** Battery pack ✓, fuse and switch in series with the supply ✓, two parallel branches ✓ each with a resistor in series with an LED ✓, LEDs the right way round, neat with labels ✓ (5)

**2.3** $\mathrm{V} = 4 \times 1{,}5 = 6\ \mathrm{V}$ ✓; capacity stays $2\ 000\ \mathrm{mAh}$ ✓; $\mathrm{t} = 2\ 000 \div 40$ ✓ $= 50\ \mathrm{h}$ ✓ (4)

**2.4** LEDs are never exactly the same; with one shared resistor the LED with the lowest voltage would take most of the current ✓, so it could burn out and the other one would be dim ✓. (2)

**2.5** About 100 mA (a little above the normal 40 mA) ✓ so that it does not blow in normal use but blows on a short circuit. (Accept 63–250 mA with a reason.) (1)

**2.6** Any TWO ✓✓: the longer leg is the anode; the flat side of the rim is the cathode; the larger plate inside is usually the cathode; test with the multimeter diode setting (it lights / conducts when red is on the anode). (2)

**[20]**

## QUESTION 3

**3.1** Connect the battery + and − to the long power rails ✓; place each resistor and LED so that they share a short row (the five holes in a row are connected) ✓; link each branch to the + and − rails with jumper wires ✓; keep the LED legs in different rows so they are not shorted, and check polarity ✓. (4)

**3.2** In order ✓✓✓✓✓: clean the board and component leads; insert the component and bend the leads slightly; heat the pad and the lead together with a clean, tinned iron; feed solder onto the joint (not the iron) until it flows; remove the solder, then the iron, and let it cool without moving; trim the leads. (5)

**3.3** Good: shiny, smooth, cone / volcano-shaped and covering the pad ✓. Dry: dull, grainy or a ball that does not wet the pad, may have cracks ✓. (2)

**3.4** Setting ✓ and connection ✓ for each: (a) DC volts on a range above 6 V, probes across (in parallel with) the battery; (b) DC mA range, break the circuit and connect the meter in series with the LED; (c) ohms range, remove the resistor (or switch off the circuit) and put the probes across it. (6)

**3.5** Any THREE ✓✓✓: LED in the wrong way round; dry joint or broken track; wrong resistor value (too high); faulty / blown LED; loose wire in the breadboard; flat battery or blown fuse (if both LEDs are off). (3)

**[20]**

## QUESTION 4

**4.1.1** $Y = T + A \cdot B$ ✓✓ (2)

**4.1.2** TAB → Y: 000→0, 001→0, 010→0, 011→1, 100→1, 101→1, 110→1, 111→1 (any consistent column order). 8 rows ✓; T = 1 rows all 1 ✓; 011 = 1 ✓; other T = 0 rows 0 ✓ (4)

**4.1.3** AND gate with inputs A and B ✓; OR gate with inputs T and the AND output ✓; output labelled Y ✓; correct symbols, neat ✓ (4)

**4.2** (a) 0111 ✓✓ (b) 3 → 0011, 9 → 1001, so 0011 1001 ✓✓ (4)

**4.3** Any TWO ✓✓: electronic switches have only two states (ON / OFF); two levels are easy to tell apart, so there are fewer errors from noise; simple, reliable circuits; easy to store and process. (2)

**4.4** Triangle with a small circle at the output ✓; truth table 0→1, 1→0 ✓ (2)

**4.5** Without it the input "floats" and can read random HIGH or LOW values from noise ✓; the pull-down resistor holds the input at 0 V (LOW) until the button connects it to the supply ✓. (2)

**[20]**

**TOTAL SECTION B: [60]**

**TOTAL: 80**

**END OF MARKING GUIDELINE**